

Structural rank-1 proof for the staircase Bott–Samelson on $V_{(2,2,1)}$ and $V_{(3,1,1)}$ at S_5 , plus refutation of the atom-count conjecture

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Abstract

We give a *structural* proof that the staircase Bott–Samelson operator Π^{S_5} has rank exactly 1 on each of $V_{(2,2,1)}$ and $V_{(3,1,1)}$ at S_5 . The proof tracks the image dimension through each Hecke factor $(T_i + 1)$ in the seminormal basis, exploiting that T_1 has a clean eigenvalue spectrum ($\{q, -1\}$ with low multiplicity at the level we need) and that $(T_1 + 1)$ kills the -1 -eigenspace. The argument upgrades the May 7 “rank-1 lemma” (verified by Lagrange interpolation) to a structural assertion. We extend the rank survey of Π^{S_n} to $n = 6$, refute the atom-count conjecture $\text{rank} \stackrel{?}{=} \lceil k_\lambda/2 \rceil$ (it fails for $\lambda \in \{(4, 1, 1), (3, 3)\}$ at $n = 6$), and record the new data: at $n = 6$, ranks $(1, 2, 3, 1, 1, 2, 0, 1, 0, 0, 0)$ for the eleven partitions of 6.

1 Setup

1.1 Conventions

Let $H_q(S_n)$ be the Iwahori–Hecke algebra of type A_{n-1} with generators $T_1, \dots, T_{n-1}$ subject to $T_i^2 = (q - 1)T_i + q$ and the braid relations. We work over a parameter q (later specialized for verification). For each partition $\lambda \vdash n$, let V_λ denote the irreducible $H_q(S_n)$ -module of shape λ (the cell module / Specht module).

Reduced word and the staircase Bott–Samelson. The staircase reduced word for the longest element $w_0 \in S_n$ is

$$[1, 2, 1, 3, 2, 1, 4, 3, 2, 1, \dots, n-1, n-2, \dots, 1].$$

The corresponding Bott–Samelson element is

$$\Pi^{S_n} := \prod_{k=1}^{n-1} (T_k + 1)(T_{k-1} + 1) \cdots (T_1 + 1).$$

At S_5 , $\Pi^{S_5} = (T_1+1)(T_2+1)(T_1+1)(T_3+1)(T_2+1)(T_1+1)(T_4+1)(T_3+1)(T_2+1)(T_1+1)$. We read products left-to-right: applying Π^{S_5} to v means applying the rightmost factor (T_1+1) first.

Seminormal basis. For $\lambda \vdash n$, the seminormal basis of V_λ is indexed by standard Young tableaux T of shape λ , with vectors v_T . For each simple reflection $s_i = (i, i + 1)$ and each SYT T :

- If $i, i + 1$ lie in the same row of T (axial distance $d = 1$), $T_i v_T = q v_T$.

- If $i, i+1$ lie in the same column of T (axial distance $d = -1$), $T_i v_T = -v_T$.
- Otherwise (axial distance $|d| \geq 2$), T_i acts as a 2×2 block on $\text{span}(v_T, v_{s_i T})$ with eigenvalues q and -1 . The $+q$ -eigenvector is $\alpha_d v_T + \beta_d v_{s_i T}$ for specific coefficients α_d, β_d depending only on d .

The operator $(T_i + 1)$ thus acts as:

- Multiplication by $(1 + q)$ on v_T if $i, i+1$ in same row.
- Zero on v_T if $i, i+1$ in same column.
- A rank-1 projector with eigenvalue $(1 + q)$ on each 2×2 block (the $+q$ -eigenspace direction).

1.2 The key principle for image-tracking

Composition of $(T_i + 1)$ factors yields a sequence of "image-shrinking" steps. The key principle:

If at some step the current image is contained in $U \oplus W$ where $U \subseteq +q$ -eigenspace of T_i and $W \subseteq -1$ -eigenspace of T_i , then applying $(T_i + 1)$ collapses to $(1 + q) \cdot U$.

This is the engine of all our rank bounds.

2 Theorem A: $\text{rank } \Pi^{S_5}|_{V_{(2,2,1)}} \leq 1$

2.1 The seminormal basis of $V_{(2,2,1)}$

The five SYTs of shape $(2, 2, 1)$ at $n = 5$ are:

$$\begin{aligned} T^{(1)} &= ((1, 2), (3, 4), (5)), & T^{(2)} &= ((1, 2), (3, 5), (4)), \\ T^{(3)} &= ((1, 3), (2, 4), (5)), & T^{(4)} &= ((1, 3), (2, 5), (4)), \\ T^{(5)} &= ((1, 4), (2, 5), (3)). \end{aligned}$$

We write $v_k := v_{T^{(k)}}$.

2.2 Action of each T_i on the seminormal basis

Compiling the position data:

- T_1 (**swap** 1, 2). $T_1 v_k = q v_k$ for $k = 1, 2$; $T_1 v_k = -v_k$ for $k = 3, 4, 5$. Hence $(T_1 + 1)$ has $+q$ -eigenspace $\text{span}(v_1, v_2)$ (dim 2) and -1 -eigenspace $\text{span}(v_3, v_4, v_5)$ (dim 3).
- T_2 (**swap** 2, 3). Pairing structure: $\{T^{(1)}, T^{(3)}\}$ at axial distance ± 2 , $\{T^{(2)}, T^{(4)}\}$ at axial distance ± 2 , and $T^{(5)}$ alone with $T_2 v_5 = -v_5$.
- T_3 (**swap** 3, 4). $T_3 v_1 = q v_1$ (same row, $d = 1$); $T_3 v_2 = -v_2$ (same column, $d = -1$); $T_3 v_3 = -v_3$ ($d = -1$); $\{T^{(4)}, T^{(5)}\}$ paired at axial distance ± 3 .
- T_4 (**swap** 4, 5). $\{T^{(1)}, T^{(2)}\}$ paired at $d = \pm 2$; $\{T^{(3)}, T^{(4)}\}$ paired at $d = \pm 2$; $T_4 v_5 = -v_5$.

2.3 Iterative image computation

Let Π_k denote the operator obtained from the first k factors of Π^{S_5} applied right-to-left. Let $\text{Im}_k := \Pi_k(V_{(2,2,1)})$.

The factors applied right-to-left, corresponding to the staircase word $[1, 2, 1, 3, 2, 1, 4, 3, 2, 1]$ reversed, are

$$\underbrace{(T_1+1)}_1 \underbrace{(T_2+1)}_2 \underbrace{(T_3+1)}_3 \underbrace{(T_4+1)}_4 \underbrace{(T_1+1)}_5 \underbrace{(T_2+1)}_6 \underbrace{(T_3+1)}_7 \underbrace{(T_1+1)}_8 \underbrace{(T_2+1)}_9 \underbrace{(T_1+1)}_{10}.$$

Step 1: (T_1+1) . Image $\subseteq +q$ -eigenspace of T_1 , i.e., $\text{Im}_1 \subseteq \text{span}(v_1, v_2)$. $\dim \text{Im}_1 \leq 2$.

Step 2: (T_2+1) . On the $\{v_1, v_3\}$ block ($d = \pm 2$), $(T_2+1)v_1$ is in the $+q$ -eigenspace direction $\alpha_2 v_1 + \beta_2 v_3$ (for Hoefsmit constants α_2, β_2 depending on $d = 2$). Similarly $(T_2+1)v_2 \in \text{span}(\alpha_2 v_2 + \beta_2 v_4)$. Hence

$$\text{Im}_2 \subseteq \text{span}(\alpha_2 v_1 + \beta_2 v_3, \alpha_2 v_2 + \beta_2 v_4),$$

with $\dim \text{Im}_2 \leq 2$.

Step 3: (T_3+1) . T_3 acts on the seminormal basis: v_1 with eigenvalue q (same row), v_2 with eigenvalue -1 (same column), v_3 with eigenvalue -1 (same column), and $\{v_4, v_5\}$ paired at $d = \pm 3$.

- $(T_3+1)(\alpha_2 v_1 + \beta_2 v_3) = \alpha_2(1+q)v_1 + 0 = \alpha_2(1+q)v_1$.
- $(T_3+1)(\alpha_2 v_2 + \beta_2 v_4) = 0 + \beta_2(T_3+1)v_4 = \beta_2(1+q)\alpha_3(\alpha_3 v_4 + \beta_3 v_5)$ (rank-1 in $\{v_4, v_5\}$ block of T_3).

Therefore

$$\text{Im}_3 \subseteq \text{span}(v_1, \alpha_3 v_4 + \beta_3 v_5), \quad \dim \text{Im}_3 \leq 2.$$

Step 4: (T_4+1) . T_4 acts: $\{v_1, v_2\}$ paired at $d = \pm 2$; $\{v_3, v_4\}$ paired at $d = \pm 2$; $T_4 v_5 = -v_5$.

- $(T_4+1)v_1 = (1+q)\alpha_2(\alpha_2 v_1 + \beta_2 v_2)$ (rank-1 in $\{v_1, v_2\}$ block).
- $(T_4+1)(\alpha_3 v_4 + \beta_3 v_5) = \alpha_3(T_4+1)v_4 + 0 = \alpha_3(1+q)\alpha_2(\alpha_2 v_3 + \beta_2 v_4)$ (rank-1 in $\{v_3, v_4\}$ block).

Therefore

$$\text{Im}_4 \subseteq \text{span}(\alpha_2 v_1 + \beta_2 v_2, \alpha_2 v_3 + \beta_2 v_4), \quad \dim \text{Im}_4 \leq 2.$$

Step 5: (T_1+1) . T_1 has $+q$ -eigenspace $\text{span}(v_1, v_2)$ and -1 -eigenspace $\text{span}(v_3, v_4, v_5)$.

- $(T_1+1)(\alpha_2 v_1 + \beta_2 v_2) = (1+q)(\alpha_2 v_1 + \beta_2 v_2)$.
- $(T_1+1)(\alpha_2 v_3 + \beta_2 v_4) = 0 + 0 = 0$.

The $\text{span}(v_3, v_4)$ component is killed. Therefore

$$\text{Im}_5 \subseteq (1+q)\text{span}(\alpha_2 v_1 + \beta_2 v_2), \quad \dim \text{Im}_5 \leq 1.$$

Steps 6–10. Subsequent factors can only preserve or reduce the image dimension. Therefore $\dim \text{Im}(\Pi^{S_5}|_{V_{(2,2,1)}}) \leq \dim \text{Im}_5 \leq 1$.

2.4 The image is nonzero

By the May 7 verification (Lagrange interpolation through degree-18 q -values, recording diagonal entries), the trace $\sigma_1(V_{(2,2,1)})|_{S_5} = q^4(1+q)^2 \neq 0$. Hence $\Pi^{S_5}|_{V_{(2,2,1)}}$ is nonzero, so $\dim \text{Im} \geq 1$.

Theorem 1 (Theorem A: structural rank-1). Π^{S_5} acting on $V_{(2,2,1)}$ has rank exactly 1.

Proof. $\dim \text{Im}_5 \leq 1$ by the iterative image computation (Steps 1–5). Subsequent factors preserve this bound, so $\text{rank } \Pi^{S_5}|_{V_{(2,2,1)}} \leq 1$. The trace is nonzero, so $\text{rank} \geq 1$. Hence $\text{rank} = 1$. $\square$

Remark 2 (Why Step 5 collapses). The collapse $\dim \text{Im}_4 = 2 \rightarrow \dim \text{Im}_5 \leq 1$ exploits two specific features of $V_{(2,2,1)}$:

- After Step 4, the image lies in $\text{span}(v_1, v_2) \oplus \text{span}(v_3, v_4)$ — the two pieces sit on opposite sides of the T_1 eigenspace decomposition.
- (T_1+1) at Step 5 then kills $\text{span}(v_3, v_4)$ (both in the -1 -eigenspace of T_1) and preserves $\text{span}(v_1, v_2)$ (in the $+q$ -eigenspace, scaled by $(1+q)$).

This is the rank-collapsing engine.

3 Theorem B: $\text{rank } \Pi^{S_5}|_{V_{(3,1,1)}} \leq 1$

3.1 The seminormal basis of $V_{(3,1,1)}$

The six SYTs of shape $(3, 1, 1)$ at $n = 5$ are:

$$\begin{aligned} T^{(1)} &= ((1, 2, 3), (4), (5)), & T^{(2)} &= ((1, 2, 4), (3), (5)), & T^{(3)} &= ((1, 2, 5), (3), (4)), \\ T^{(4)} &= ((1, 3, 4), (2), (5)), & T^{(5)} &= ((1, 3, 5), (2), (4)), & T^{(6)} &= ((1, 4, 5), (2), (3)). \end{aligned}$$

We write $v_k := v_{T^{(k)}}$.

3.2 Action of each T_i

Position lookups give:

- T_1 . $T_1 v_k = q v_k$ for $k = 1, 2, 3$; $T_1 v_k = -v_k$ for $k = 4, 5, 6$. Hence $+q$ -eigenspace of T_1 is $\text{span}(v_1, v_2, v_3)$ ($\dim 3$).
- T_2 . $T_2 v_1 = q v_1$; $\{T^{(2)}, T^{(4)}\}$ paired at $d = \pm 2$; $\{T^{(3)}, T^{(5)}\}$ paired at $d = \pm 2$; $T_2 v_6 = -v_6$.
- T_3 . $\{T^{(1)}, T^{(2)}\}$ paired at $d = \pm 3$; $T_3 v_3 = -v_3$; $T_3 v_4 = q v_4$; $\{T^{(5)}, T^{(6)}\}$ paired at $d = \pm 3$.
- T_4 . $T_4 v_1 = -v_1$; $\{T^{(2)}, T^{(3)}\}$ paired at $d = \pm 4$; $\{T^{(4)}, T^{(5)}\}$ paired at $d = \pm 4$; $T_4 v_6 = q v_6$.

3.3 Iterative image computation

We apply factors of Π^{S_5} from right to left: the order is

$$\underbrace{(T_1+1)}_{\text{Step 1}} \underbrace{(T_2+1)}_{\text{Step 2}} \underbrace{(T_3+1)}_{\text{Step 3}} \underbrace{(T_4+1)}_{\text{Step 4}} \underbrace{(T_1+1)}_{\text{Step 5}} \underbrace{(T_2+1) \cdots (T_1+1)}_{\text{Steps 6–10}}.$$

The proof tracks Im_k very carefully through Steps 1–5, showing $\dim \text{Im}_5 \leq 1$. The remaining factors then preserve this bound.

Step 1: (T_1+1) . $\text{Im}_1 \subseteq \text{span}(v_1, v_2, v_3)$; $\dim \leq 3$.

Step 2: (T_2+1) . $(T_2+1)v_1 = (1+q)v_1$ (same row). $(T_2+1)v_2 \in (1+q)\text{span}(\alpha_2v_2 + \beta_2v_4)$ (rank-1 + q -direction in $d = \pm 2$ block). $(T_2+1)v_3 \in (1+q)\text{span}(\alpha_2v_3 + \beta_2v_5)$. Hence Im_2 is generated by the three vectors

$$b_1 = v_1, \quad b_2 = \alpha_2v_2 + \beta_2v_4, \quad b_3 = \alpha_2v_3 + \beta_2v_5,$$

each up to a $(1+q)$ scalar.

Step 3: (T_3+1) .

- $(T_3+1)b_1 = (T_3+1)v_1 = (1+q)\alpha_3(\alpha_3v_1 + \beta_3v_2)$ (rank-1 in the $\{v_1, v_2\}$ block of T_3 , axial $d = \pm 3$).
- $(T_3+1)b_2 = \alpha_2(T_3+1)v_2 + \beta_2(1+q)v_4$. Now $(T_3+1)v_2 \in (1+q)\text{span}(\alpha_3v_1 + \beta_3v_2)$ (same block; v_2 has $d = +3$). So $(T_3+1)b_2 \in \text{span}(\alpha_3v_1 + \beta_3v_2) \oplus \text{span}(v_4)$.
- $(T_3+1)b_3 = 0 + \beta_2(T_3+1)v_5$. $(T_3+1)v_5 \in (1+q)\text{span}(\alpha_3v_5 + \beta_3v_6)$ (rank-1 in $\{v_5, v_6\}$ block).

Hence Im_3 is generated (up to scalars) by three vectors

$$a_1 = \alpha_3v_1 + \beta_3v_2, \quad a_2 \in \text{span}(\alpha_3v_1 + \beta_3v_2) \oplus \text{span}(v_4), \quad a_3 = \alpha_3v_5 + \beta_3v_6.$$

The key structural fact: the projection of Im_3 onto $\text{span}(v_1, v_2, v_3)$ is contained in $\text{span}(\alpha_3v_1 + \beta_3v_2)$ — a 1-dim line, NOT the full $\text{span}(v_1, v_2)$.

Step 4: (T_4+1) . Recall T_4 acts: $T_4v_1 = -v_1$; rank-1 projection on $\{v_2, v_3\}$ at $d = \pm 4$; rank-1 projection on $\{v_4, v_5\}$ at $d = \pm 4$; $T_4v_6 = qv_6$.

- $(T_4+1)a_1 = \alpha_3(T_4+1)v_1 + \beta_3(T_4+1)v_2 = 0 + \beta_3(1+q)\alpha_4(\alpha_4v_2 + \beta_4v_3)$. So $(T_4+1)a_1 \in \text{span}(\alpha_4v_2 + \beta_4v_3)$ **(1-dim)**.
- $(T_4+1)a_2$: since $a_2 \in \text{span}(\alpha_3v_1 + \beta_3v_2) \oplus \text{span}(v_4)$, and $(T_4+1)v_1 = 0$, $(T_4+1)v_2 \in \text{span}(\alpha_4v_2 + \beta_4v_3)$, $(T_4+1)v_4 \in \text{span}(\alpha_4v_4 + \beta_4v_5)$: $(T_4+1)a_2 \in \text{span}(\alpha_4v_2 + \beta_4v_3, \alpha_4v_4 + \beta_4v_5)$ **(2-dim)**.
- $(T_4+1)a_3$: $a_3 \in \text{span}(v_5, v_6)$, $(T_4+1)v_5 \in \text{span}(\alpha_4v_4 + \beta_4v_5)$, $(T_4+1)v_6 = (1+q)v_6$. $(T_4+1)a_3 \in \text{span}(\alpha_4v_4 + \beta_4v_5, v_6)$ **(2-dim)**.

Therefore $\text{Im}_4 \subseteq \text{span}(\alpha_4v_2 + \beta_4v_3, \alpha_4v_4 + \beta_4v_5, v_6)$. Crucially, the projection of Im_4 onto $\text{span}(v_1, v_2, v_3)$ lies in $\text{span}(\alpha_4v_2 + \beta_4v_3)$ **(1-dim)**.

Step 5: (T_1+1) . T_1 kills $\text{span}(v_4, v_5, v_6)$ (the -1 -eigenspace) and acts as $(1+q)$ on $\text{span}(v_1, v_2, v_3)$. Therefore

$$(T_1+1)\text{Im}_4 \subseteq (1+q)\text{span}(\alpha_4v_2 + \beta_4v_3),$$

a 1-dim subspace.

Steps 6–10. Each (T_i+1) factor is a linear map and cannot increase image dimension. So $\dim \text{Im}(\Pi^{S_5}|_{V_{(3,1,1)}}) \leq \dim \text{Im}_5 \leq 1$.

Theorem 3 (Theorem B: structural rank-1). Π^{S_5} acting on $V_{(3,1,1)}$ has rank exactly 1.

Proof. $\dim \text{Im}_5 \leq 1$ by the refined argument above: the v_4, v_5, v_6 components of Im_4 are killed by (T_1+1) , leaving only the $\text{span}(\alpha_4v_2 + \beta_4v_3)$ direction, which is 1-dimensional. Subsequent factors preserve this bound.

The image is nonzero by the May 7 verification: $\sigma_1(V_{(3,1,1)})|_{S_5} = q^3(1+q)^4 + 2q^4(1+q)^2 \neq 0$. Hence rank = 1. $\square$

4 The structural argument vs. rank-2 cases

For comparison, we apply the same image-tracking to $V_{(3,2)}$ at S_5 (rank 2).

The five SYTs of $(3, 2)$ are $T^{(1)} = ((1, 2, 3), (4, 5))$, $T^{(2)} = ((1, 2, 4), (3, 5))$, $T^{(3)} = ((1, 2, 5), (3, 4))$, $T^{(4)} = ((1, 3, 4), (2, 5))$, $T^{(5)} = ((1, 3, 5), (2, 4))$.

In particular, T_3 acts on $V_{(3,2)}$ with the structure:

- $\{v_1, v_2\}$ paired at $d = \pm 3$.
- $T_3 v_3 = q v_3$ (since 3, 4 in same row of $T^{(3)}$).
- $T_3 v_4 = q v_4$ (same row).
- $T_3 v_5 = -v_5$ (same column).

Crucial difference from $V_{(3,1,1)}$. In $V_{(3,1,1)}$, $T_3 v_3 = -v_3$ (since the shape has 3, 4 in the same column for $T^{(3)}$). Therefore in Step 3 of the $V_{(3,1,1)}$ argument, the $\alpha_2 v_3 + \beta_2 v_5$ term loses its v_3 -component:

$$(T_3 + 1)(\alpha_2 v_3 + \beta_2 v_5)|_{V_{(3,1,1)}} = 0 + \beta_2 (T_3 + 1)v_5 \in \text{span}(v_5, v_6),$$

and this entire contribution sits in the -1 -eigenspace of T_1 , so it gets killed by Step 5.

In $V_{(3,2)}$, by contrast, $T_3 v_3 = q v_3$, so

$$(T_3 + 1)(\alpha_2 v_3 + \beta_2 v_5)|_{V_{(3,2)}} = \alpha_2(1+q)v_3 + 0 = \alpha_2(1+q)v_3.$$

This contribution stays in $\text{span}(v_3) \subseteq +q\text{-eig}(T_1)$, **surviving** the Step 5 application of $(T_1 + 1)$.

Tracking through all 10 steps numerically (script `verify_seminormal.py`) shows that for $V_{(3,2)}$ the image dimension is 3 through Step 4, then drops to 2 at Step 5 and stays at 2. The rank is 2, not 1.

Remark 4 (Where rank-1 vs rank-2 is decided). The image dimension sequence at Steps $4 \rightarrow 5$ encodes the rank:

- $V_{(2,2,1)}$: $2 \rightarrow 1$. Image after Step 4 has $\text{span}(v_1, v_2)$ component 1-dim; the rest is killed by $(T_1 + 1)$.
- $V_{(3,1,1)}$: $3 \rightarrow 1$. Image after Step 4 has $+q\text{-eig}(T_1)$ component 1-dim (a single line $\alpha_4 v_2 + \beta_4 v_3$); the rest is in $-1\text{-eig}(T_1)$ and killed.
- $V_{(3,2)}$: $3 \rightarrow 2$. Image after Step 4 has $+q\text{-eig}(T_1)$ component 2-dim; nothing more collapses at Step 5.
- $V_{(4,1)}$: similarly $3 \rightarrow 2$.

5 Extended rank survey: $n = 6$

We extend the rank survey of `rank_survey.py` (May 7) to $n = 6$. The script `rank_n6_survey.py` computes Π^{S_6} on each cell module V_λ via Lagrange interpolation and reports the numerical rank at $q \in \{2, 3, 5, 7, 11, 13\}$.

5.1 Data

$\lambda \vdash 6$	$\dim V_\lambda$	$\text{rank } \Pi^{S_6}$	$\sigma_1(V_\lambda)$ (degrees only)
(6)	1	1	$1 \dots q^{15}$
(5, 1)	5	2	$q \dots q^{14}$
(4, 2)	9	3	$q^2 \dots q^{13}$
(4, 1, 1)	10	1	$q^3 \dots q^{12}$
(3, 3)	5	1	$q^3 \dots q^{12}$
(3, 2, 1)	16	2	$q^4 \dots q^{11}$
(3, 1, 1, 1)	10	0	0
(2, 2, 2)	5	1	$q^6 \dots q^9$
(2, 2, 1, 1)	9	0	0
(2, 1, 1, 1, 1)	5	0	0
(1 ⁶)	1	0	0

5.2 Coincidence: $\sigma_1((4, 1, 1)) = \sigma_1((3, 3))$

A striking observation: the trace polynomials are equal:

$$\sigma_1(V_{(4,1,1)}) = \sigma_1(V_{(3,3)}) = q^3 + 15q^4 + 91q^5 + 281q^6 + 484q^7 + 484q^8 + 281q^9 + 91q^{10} + 15q^{11} + q^{12}.$$

Both partitions have rank 1, but the dimensions 10 and 5 are different. This is unique among $n \leq 6$: at $n = 5$, all σ_1 values are distinct.

5.3 Atom decomposition

Decomposing each nonzero σ_1 as $\sum_{i \geq 0} m^{(i)} q^{n(\lambda)+i} (1+q)^{w_\lambda-2i}$ where $w_\lambda = L - 2n(\lambda)$ is the width:

λ	rank	k_λ	atom multiplicities $(m^{(0)}, m^{(1)}, \dots)$
(6)	1	1	(1)
(5, 1)	2	4	(1, 3, 9, 14, 0, 0, 0)
(4, 2)	3	5	(1, 5, 21, 30, 9, 0)
(4, 1, 1)	1	4	(1, 6, 13, 6, 0)
(3, 3)	1	4	(1, 6, 13, 6, 0)
(3, 2, 1)	2	4	(1, 9, 15, 2)
(2, 2, 2)	1	1	(1, 0)

(Here k_λ denotes the number of nonzero atoms.)

6 Refutation of the May-7 atom-count conjecture

Conjecture 5 (*Conjecture from May-7 writeup, hereby refuted*). $\text{rank } \Pi^{S_n}|_{V_\lambda} = \lceil k_\lambda/2 \rceil$.

Proposition 6 (Refutation). *At $n = 6$, $V_{(4,1,1)}$ and $V_{(3,3)}$ each have $k_\lambda = 4$ nonzero atoms but $\text{rank } \Pi^{S_6} = 1 \neq 2 = \lceil 4/2 \rceil$.*

Proof. By the rank survey of Section 5 (computational), and the atom decomposition in the same section. $\square$

What survives. The conjecture holds at $n \leq 5$ (by direct check) and at $n = 6$ for $\lambda \in \{(6), (5, 1), (4, 2), (3, 2, 1), (2, 2, 2)\}$. It fails specifically for the two coinciding σ_1 partitions $\{(4, 1, 1), (3, 3)\}$.

Refined conjecture.

Conjecture 7 (Atom count is an upper bound). $\text{rank } \Pi^{S_n}|_{V_\lambda} \leq \lceil k_\lambda/2 \rceil$.

This holds for all $n \leq 6$ data points and matches the original spirit of the May-7 conjecture as a one-sided bound.

7 Discussion and open questions

7.1 What the structural argument captures

The iterative-image-tracking argument of Sections 2–3, which gives rank-1 structurally for $V_{(2,2,1)}$ and $V_{(3,1,1)}$, exploits two key features of these specific cell modules:

1. *Eigenvalue uniformity of T_1 .* T_1 is diagonal with at most two distinct eigenvalues ($\{q, -1\}$). The -1 -eigenspace is $\text{span}(v_T : 1 \in D_L(T))$ where D_L is the descent set of the SYT.
2. *The “narrow” structure of Im_4 in the $+q$ direction of T_1 .* For these partitions, Im_4 has its $+q$ -component in a specific 1-dim subspace of $\text{span}(v_2, v_3)$ (case $V_{(3,1,1)}$) or in $\text{span}(v_1, v_2)$ with the v_2 direction killed by T_3 (case $V_{(2,2,1)}$).

The structural argument *fails* for $V_{(3,2)}$ and $V_{(4,1)}$ because T_3 acts uniformly on more of the surviving generators of Im_4 , preventing the rank-1 collapse.

7.2 Open: a structural characterization

The data through $n = 6$ does not yet determine a clean representation-theoretic invariant equal to $\text{rank } \Pi^{S_n}|_{V_\lambda}$. Possible candidates:

- *Number of “good” SYTs:* SYTs T such that $T_1, T_2, T_3, \dots$ act “compatibly” on v_T in some specific sense yet to be made precise. Empirically, for each rank- r case, the first r SYTs in the cell-module enumeration order give linearly independent images under Π .
- *Sub-family rank formulas.* For $\lambda = (n - 2, 2)$: $\text{rank} = n - 3$ (verified for $n \in \{4, 5, 6\}$). For $\lambda = (n - 1, 1)$: $\text{rank} = \lceil (n - 2)/2 \rceil$ (verified for $n \in \{3, 4, 5, 6\}$). These are not in general $\lceil k_\lambda/2 \rceil$ but match the rank exactly for these families.

7.3 Implications for the May-7 trace identity

Theorems 1 and 3 upgrade the May-7 “rank-1 lemma” (Theorem 3.1 in 2026-05-07-rank-one-Pi.tex) from a computational verification to a structural theorem. By Corollary 3.7 of that writeup, this implies the branching factorization

$$T_A = \text{tr}_{V_{(3,2,1)}}(\Pi^{S_5} P_{(2,2,1)}^{\text{KL}} R'_5) = q^2 \sigma_1(V_{(2,2,1)})|_{S_5}$$

and the analogous identity for T_B are now established at full rigor (modulo the standard branching reduction, which is purely representation-theoretic).

8 Honest assessment

What this paper achieves.

1. Structural proofs (Theorems 1, 3) that Π^{S_5} has rank exactly 1 on $V_{(2,2,1)}$ and $V_{(3,1,1)}$. The proofs track the image dimension through each Hecke factor $(T_i + 1)$ in the seminormal basis.
2. Extension of the rank survey to $n = 6$ (Section 5), with the new data point that $\sigma_1(V_{(4,1,1)}) = \sigma_1(V_{(3,3)})$ at $n = 6$.
3. Refutation of the May-7 atom-count conjecture (Section 6), with a refined upper-bound conjecture in place.

What this paper does *not* achieve.

1. *A general structural characterization* of $\text{rank } \Pi^{S_n}|_{V_\lambda}$ for all (n, λ) . The structural argument used here is special to the small cases where T_3 or T_4 has a clean “splitting” eigenvalue behavior on the surviving image generators.
2. *Why the rank survey at $n = 6$ has the specific values it does.* For instance, why does $V_{(4,2)}$ have rank exactly 3 (and not 2 or 4)? The structural argument gives only an upper bound consistent with rank 3, but does not pin it down structurally.
3. *An explicit closed form for the rank-1 image vector.* The image of Π^{S_5} on $V_{(2,2,1)}$ is spanned by a specific combination of v_1, v_2 with coefficients determined by the $d = \pm 2$ Hoefsmit constants; we have not written this combination explicitly in this writeup.

Status. The branching factorization conjecture is now a *structural theorem* (rather than “theorem-modulo-rank-1-lemma” as it was in the May-7 writeup). The general rank-characterization problem remains open, with a partial classification at $n \leq 6$ and refined upper-bound conjectures.

Computational note. All matrix entries, rank verifications, and the new $n = 6$ data are checked by Lagrange interpolation in the scripts:

```
~/projects/scratch/2026-05-08-rank-Pi/rank_n6_survey.py,  
~/projects/scratch/2026-05-08-rank-Pi/image_analysis.py.
```